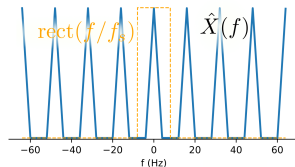


Reconstruction

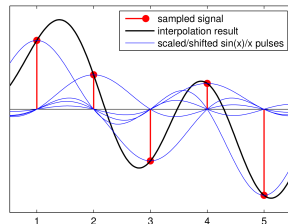
- 1 The image centered on the origin is identical with the original $X(f)$ spectrum.
- 2 We use a rectangle function to “cut” out that one image from $\hat{X}(f)$. This is known as **reconstruction filtering**.
- 3 We can reconstruct the original $x(t)$ signal by applying an inverse F.T. on this image.



$$X(f) = \hat{X}(f) \cdot \text{rect}(f/f_s) = \mathcal{F} \left[\hat{x} * \frac{1}{T} \text{sinc}_\pi(t/T) \right]$$

The reconstructed signal becomes:

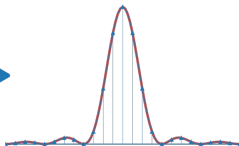
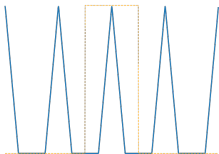
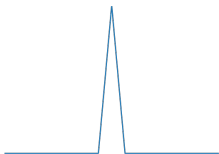
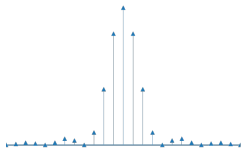
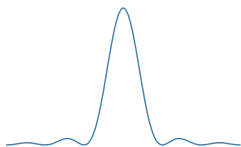
$$\begin{aligned} x(t) &= \mathcal{F}^{-1}[X](t) = \hat{x} * \frac{1}{T} \text{sinc}_\pi(t/T) = \\ &= \sum_{n=-\infty}^{\infty} x[n] \frac{\sin \pi(t - nT)/T}{\pi(t - nT)/T} . \end{aligned}$$



The original signal can be reconstructed by linearly combining sinc_π functions centered on each sample point nT (by interpolating the sampled signal $x[n] = x(nT)$ back into the continuous waveform).

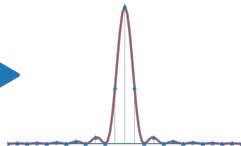
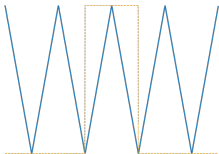
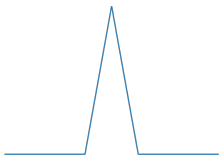
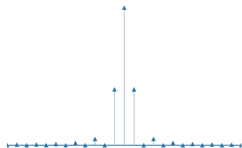
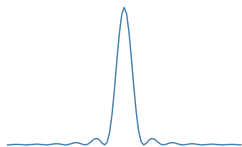
Reconstruction example: functioning case

$$f_{\max} < \frac{f_s}{2}$$



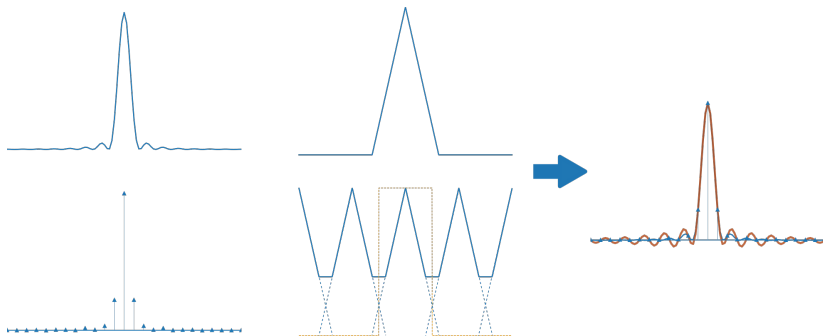
Reconstruction example: borderline case

$$f_{\max} = \frac{f_s}{2}$$



Reconstruction example: failure

$$f_{\max} > \frac{fs}{2}$$

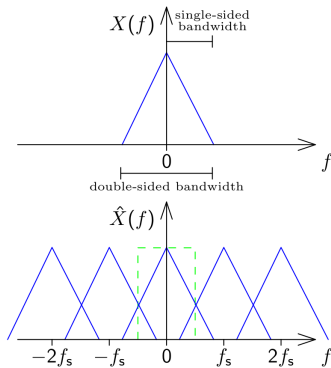


After applying the reconstruction filter we do not recover the original spectrum.

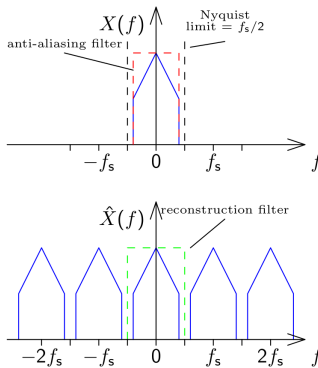
Antialiasing filter: illustration

If we do not want to sample at sufficiently high rate so that the entire band is covered (because of performance considerations and/or lack of interest in the higher frequencies) and still want to have control in the reconstruction process we first apply an antialiasing filter. Otherwise higher than $f_s/2$ frequency components will appear in the digitized data as low frequency spurious signals:

Without anti-aliasing filter



With anti-aliasing filter



original
continuous
signal



antialiasing
filter



sampling



reconstruction
filter



reconstructed
signal

Step 1 - Antialiasing filter:

$$x(t) \xrightarrow{\text{filtering}} \tilde{x}(t) = \mathcal{F}^{-1}[\tilde{X}(f)] = x(t) * T \operatorname{sinc}_{\pi}(t/T), \quad \text{where} \quad \tilde{X}(f) = X(f) \cdot \operatorname{rect}(f/f_s)$$

Step 2 - Sampling: similarly as before with $x(t)$, i.e., using the Dirac-comb and obtaining a $\hat{\tilde{x}}(t)$ signal (corresponding to the $\tilde{x}[n]$ sequence) that has a $f_s = 1/T$ periodic $\hat{\tilde{X}}(f)$ spectrum with the “clean” images of $\tilde{X}(f)$.

Step 3 - Reconstruction filter:

$$\begin{aligned} \tilde{x}(t) &= \mathcal{F}^{-1}[\tilde{X}](t) = \hat{\tilde{x}} * \frac{1}{T} \operatorname{sinc}_{\pi}(t/T) = \\ &= \sum_{n=-\infty}^{\infty} \tilde{x}[n] \frac{\sin \pi(t - nT)/T}{\pi(t - nT)/T} . \end{aligned}$$

→ so we reconstructed not the original signal but the one subject to the antialias filter.

Exercises

Starting from the definitions:

$$X(f) \equiv \int_{-\infty}^{\infty} x(t) e^{2\pi i f t} dt, \quad x(t) = \int_{-\infty}^{\infty} X(f) e^{-2\pi i f t} df, \quad \hat{x}(t) = x(t) \cdot \Pi_T(t), \quad x[n] = x(nT).$$

Show that

$$\textcircled{1} \quad \int_{-\infty}^{\infty} e^{2\pi i f t} df = \delta(t)$$

$$\textcircled{2} \quad \mathcal{F}[\delta(t - \tau)] = e^{2\pi i f \tau}$$

$$\textcircled{3} \quad \mathcal{F}[\text{rect}(t)] = \text{sinc}(f)$$

$$\textcircled{4} \quad \int_0^{\infty} \frac{\sin \alpha t}{t} dt = \pi \delta(\alpha)$$

Hint: Make use of $d/d\alpha \left[\int_0^{\infty} \sin(\alpha t)/t dt \right] = \int_0^{\infty} \cos(\alpha t) dt$

$$\textcircled{5} \quad \mathcal{F}[\text{sinc}(t)] = \text{rect}(f)$$

$$\textcircled{6} \quad \int_{-\infty}^{\infty} \text{rect}(t) dt = \int_{-\infty}^{\infty} \text{sinc}(t) dt = 1$$

$$\textcircled{7} \quad x(t) \cdot y(t) = \mathcal{F}^{-1}[X * Y]$$

$$\textcircled{8} \quad \mathcal{F}[\text{rect}(t/T)] = T \text{sinc}(fT)$$

$$\textcircled{9} \quad [x * y](t) = [y * x](t).$$

$$\textcircled{10} \quad \mathcal{F}[x * y](f) = X(f) \cdot Y(f).$$

$$\textcircled{11} \quad \text{sinc}(m - n) = \delta_{mn}$$

$$\textcircled{12} \quad \hat{X}(f) \equiv \mathcal{F}[\hat{x}(t)](f) = \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

$$\textcircled{13} \quad \bar{x}(t) \equiv \mathcal{F}^{-1}[\hat{X}(f) \cdot \text{rect}(f/f_s)](t) =$$

$$= \sum_{n=-\infty}^{\infty} x[n] \frac{\sin \pi(t - nT)}{\pi(t - nT)}$$

$$\textcircled{14} \quad \text{Argue that } \bar{x}(t) \text{ is identical with } x(t) \text{ if the latter is bandwidth limited to the } [-f_s/2, f_s/2] \text{ interval.}$$

Summary: Sampling and reconstruction

$x(t)$ continuous signal with Fourier transform $X(f)$ is sampled with sampling rate $f_s = 1/T$

$$x[n] = x(nT) = \int_{-\infty}^{\infty} x(\tau) \delta(nT - \tau) d\tau = x(t) * \delta(t - nT)$$

The continuous representation of the sampled signal is

$$\hat{x}(t) = x(t) \cdot \Pi_T(t) = \mathcal{F}^{-1} \left[X * \frac{1}{f_s} \Pi_{f_s} \right]$$

where

$$\Pi_T(t) = T \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$\mathcal{F} [\Pi_T(t)](f) = \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = \frac{1}{f_s} \Pi_{f_s}(f)$$

$$\hat{X}(f) = \mathcal{F}[\hat{x}(t)](f) = T \sum_{n=-\infty}^{\infty} x[n] e^{2\pi i n \frac{f}{f_s}}$$

which is the Fourier series expansion of an f_s periodic function where $x[n]$ are the Fourier coefficients also defined as

$$x[n] = \int_{-f_s/2}^{f_s/2} \hat{X}(f) e^{-2\pi i n \frac{f}{f_s}} df =$$

$$\begin{aligned} \text{Indeed } x[n] &= \int_{-\infty}^{\infty} \text{rect}(f/f_s) \hat{X}(f) e^{-2\pi i n \frac{f}{f_s}} df = \\ &= \mathcal{F}^{-1} [\text{rect}(f/f_s) \hat{X}(f)](nT) = \left[\frac{1}{T} \text{sinc}_{\pi}(t/T) * \hat{x}(t) \right](nT) = \\ &= \frac{1}{T} \int_{-\infty}^{\infty} x(\tau) \Pi_T(\tau) \text{sinc}_{\pi} \left(\frac{nT - \tau}{T} \right) d\tau = \sum_{m=-\infty}^{\infty} x[m] \text{sinc}_{\pi}(n-m) = \\ &= \sum_{m=-\infty}^{\infty} x[m] \delta_{mn} = x[n] \end{aligned}$$

where

$$\text{rect}(t) = \begin{cases} 1, & |t| < 1/2 \\ 0, & |t| > 1/2 \\ \frac{1}{2}, & |t| = 1/2 \end{cases} = \mathcal{F}(\text{sinc}_{\pi}(f))(t)$$

$$\text{sinc}_{\pi}(f) = \frac{\sin(\pi f)}{\pi f} = \mathcal{F}(\text{rect}(t))(f)$$

$$\mathcal{F} [\text{rect}(t/T)](f) = \frac{1}{f_s} \text{sinc}_{\pi}(f/f_s) \text{ or}$$

$$\mathcal{F} [\text{rect}(f/f_s)](t) = \frac{1}{T} \text{sinc}_{\pi}(t/T)$$



Alternatively

$$\begin{aligned} \hat{X}(f) &= \mathcal{F}[\hat{x}(t)](f) = \mathcal{F}[x(t) \cdot \Pi_T(t)] = \\ &= X * \frac{1}{f_s} \Pi_{f_s} = \frac{1}{f_s} \int_{-\infty}^{\infty} X(\nu) \Pi_{f_s}(f - \nu) d\nu = \\ &= \sum_{n=-\infty}^{\infty} X(f - nf_s) \end{aligned}$$

If the support (region where it is different from zero) of $X(f)$ is less than f_s then $\hat{X}(f)$ will be a shifted set of copies of the original $X(f)$ spectrum. In this case "cutting out" a $-f_s/2, f_s/2$ interval from $\hat{X}(f)$, yielding $\tilde{X}(f)$, will recover the original $X(f)$. We do the cutting by the $\text{rect}(f/f_s)$ function:

$$\tilde{X}(f) = \hat{X}(f) \cdot \text{rect}(f/f_s) = \mathcal{F} \left[\hat{x} * \frac{1}{T} \text{sinc}_{\pi}(t/T) \right]$$

The reconstructed signal becomes:

$$\tilde{x}(t) = \mathcal{F}^{-1}[\tilde{X}](t) = \hat{x} * \frac{1}{T} \text{sinc}_{\pi}(t/T) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin \pi(t-nT)/T}{\pi(t-nT)/T}$$

Continuous vs discrete Fourier transform

Let's start from the continuous aperiodic signal $x(t)$.

We sample with sampling period $T = 1/f_s$:

$$\hat{x}(t) = x(t) \cdot \Pi_T(t) = \sum_n x[n] \delta(t - nT)$$

1 an infinite set of equidistant Dirac-deltas at T intervals.

$$\hat{X}(f) = \mathcal{F}(\hat{x}(t))(f) = T \sum_{n=-\infty}^{\infty} x[n] e^{2\pi i n \frac{f}{f_s}} \quad \text{periodic spectrum } f \in \left[-\frac{f_s}{2}, \frac{f_s}{2}\right]$$

$\hat{X}(f)$ continuous and periodic, $x[n]$ sampled and aperiodic.

If $x(t)$ is periodic with period $NT \rightarrow \hat{x}(t)$ is also periodic $\rightarrow$ the latter's Fourier-transform becomes (see dsp03.pdf)

$$\hat{X}(f) = \sum_{k=-\infty}^{+\infty} c_k \delta\left(f - \frac{k}{(NT)}\right),$$

an infinite set of equidistant Dirac-deltas at f_s/N away from one another $\rightarrow$ a sampled spectrum.

① Sampled signal $\xrightarrow{\mathcal{F}}$ periodic spectrum

② Periodic signal $\xrightarrow{\mathcal{F}}$ sampled spectrum

Sampled & Periodic signal $\xrightarrow{\mathcal{F}}$ sampled & periodic spectrum

Assume a periodic signal $\rightarrow x(t + NT) = x(t) \rightarrow x[n] = x[n + N]$.

$$\begin{aligned}\hat{X}(f) &= T \sum_n x[n] e^{2\pi i n f T} = T \sum_n x[n + N] e^{2\pi i n f T} = \\ &= T \sum_n x[n] e^{2\pi i (n-N) f T} = \hat{X}(f) e^{-2\pi i N f T}\end{aligned}$$

$$\rightarrow N f T = k \in \mathbb{Z} \rightarrow f = \frac{k}{N} f_s.$$

In the same time because of the f_s periodicity of $\hat{X}(f)$ it is sufficient to reduce f to $[-f_s/2, f_s/2)$, i.e., $k \in \{-\frac{N}{2}, \dots, \frac{N}{2} - 1\}$.

Assuming the usage of anti-aliasing filter (no frequencies beyond $f_s/2$)

$$\hat{X}(f) = X(f) \text{ if } f \in [-f_s/2, f_s/2)$$

$$\hat{X}(f) = X\left(\frac{k}{N} f_s\right) \equiv X[k] = T \sum_{n=-N/2}^{N/2-1} x[n] e^{2\pi i n k / N}$$

The inverse Fourier-transform ($x \rightarrow X$, $T \rightarrow f_s/N$, sign change in the exponential):

$$x[n] = \frac{f_s}{N} \sum_{k=-N/2}^{N/2-1} X[k] e^{-2\pi i n k / N}$$

If rescale the Fourier-transform $X[m]$ to $X[k]/T$ then the two transforms:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{2\pi i}{N} n k}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{\frac{2\pi i}{N} n k}$$

Discrete Fourier transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{2\pi i}{N} nk}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{\frac{2\pi i}{N} nk}$$

The $1/N$ factor is required to ensure that the two transforms are each other's inverse.

$$\begin{aligned} x[n] &= \frac{1}{N} \sum_k \sum_m x[m] e^{-\frac{2\pi i}{N} mk} e^{\frac{2\pi i}{N} nk} \\ &= \frac{1}{N} \sum_m x[m] \sum_k e^{\frac{2\pi i}{N} (m-n)k} = \frac{1}{N} \sum_m x[m] N \delta_{nm} = x[n] \end{aligned}$$

because $\sum_{k=0}^{N-1} e^{\frac{2\pi i}{N} (m-n)k} = \begin{cases} N, & m = n \\ 0, & m \neq n \end{cases}$

The number of input values (time points) are the same as the number of output values (frequency points).

Exercise

Show that

$$\sum_{k=0}^{N-1} e^{\frac{2\pi i}{N} (m-n)k} = \begin{cases} N, & m = n \\ 0, & m \neq n \end{cases}$$

Hint Separately consider the $m = n$ and the $m \neq n$ cases.

Method 1: View the sum as a geometric one (sum of k th powers of an exponential)

Method 2: Split the sum into two halves, i.e. 0 to $N/2 - 1$ and $N/2$ to $N - 1$ then perform the change of variable $k \rightarrow k - N/2$ in the second sum.

Method 3: Draw the terms in the complex plane for $m - n = 1$ and study the sum geometrically. What changes if $n \notin \{0, 1\}$?